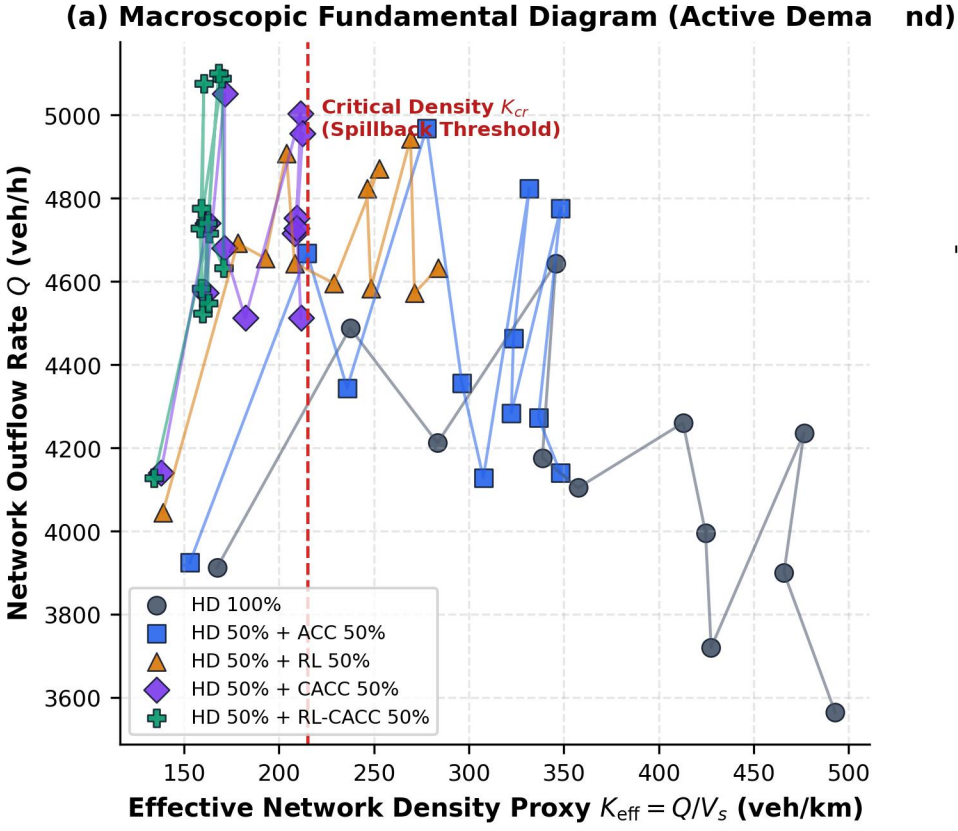
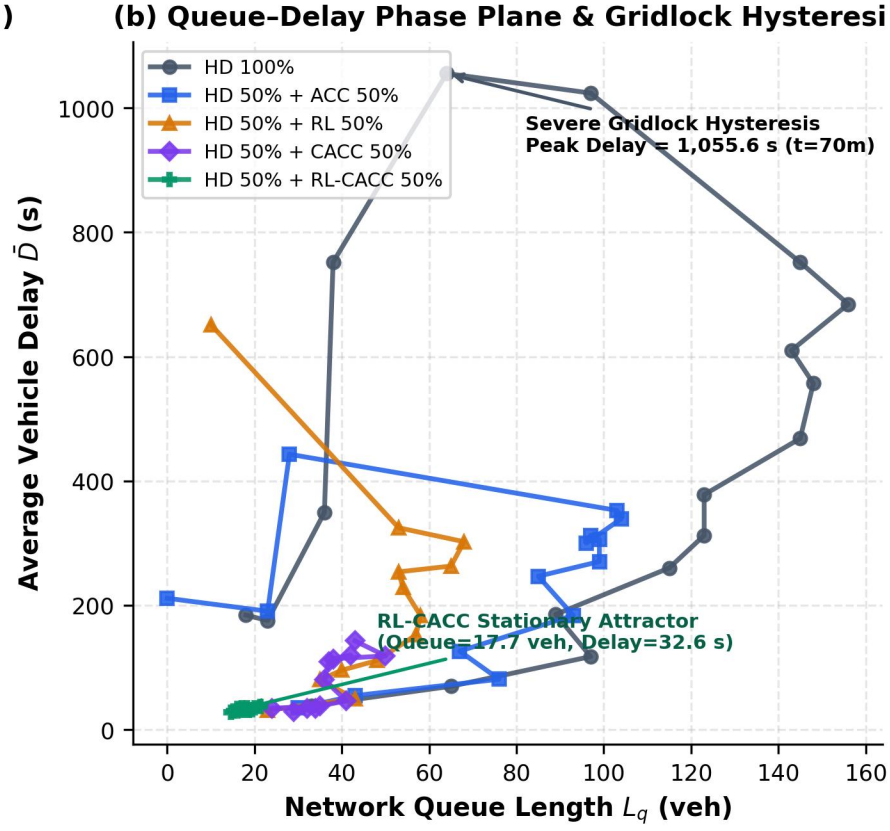
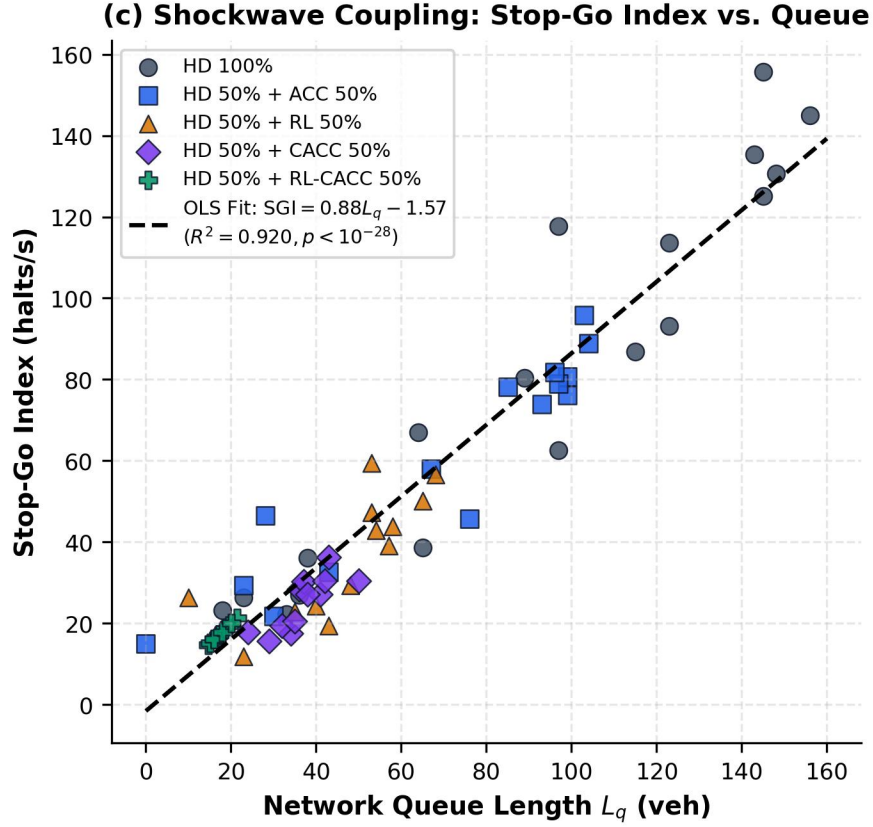

(a) Macroscopic Fundamental Diagram (Active Dema nd)
Critical Density K_{cr}
(Spillback Threshold)
HD 100%
HD 50% + ACC 50%
HD 50% + RL 50%
HD 50% + CACC 50%
HD 50% + RL-CACC 50%
Network Outflow Rate Q (veh/h)
Effective Network Density Proxy $K_{\text{eff}} = Q/V_s$ (veh/km)


(b) Queue–Delay Phase Plane & Gridlock Hysteresi
HD 100%
HD 50% + ACC 50%
HD 50% + RL 50%
HD 50% + CACC 50%
HD 50% + RL-CACC 50%
Severe Gridlock Hysteresis
Peak Delay = 1,055.6 s (t=70m)
RL-CACC Stationary Attractor
(Queue=17.7 veh, Delay=32.6 s)
Average Vehicle Delay $\bar{D}$ (s)
Network Queue Length L_q (veh)

s


(c) Shockwave Coupling: Stop-Go Index vs. Queue
HD 100%
HD 50% + ACC 50%
HD 50% + RL 50%
HD 50% + CACC 50%
HD 50% + RL-CACC 50%
OLS Fit: SGI = 0.88L_q − 1.57
($R^2 = 0.920, p < 10^{-28}$)
Stop-Go Index (halts/s)
Network Queue Length L_q (veh)